

STOLNICI, Romania

WMO: 154160

Lat: 44.5628N

Lon: 24.7897E

Elev: 689

StdP: 14.33

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.7	13.8	7.0	8.1	10.3	11.4	10.1	15.7	29.8	33.7	23.0	33.5	1.9	330	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.1	91.9	69.1	89.5	68.5	86.7	68.1	72.6	86.1	71.2	83.9	69.9	82.2	4.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.5	107.5	77.1	67.2	102.5	75.8	65.9	98.0	74.6	36.6	85.9	35.4	84.1	34.3	82.6	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.0	16.6	14.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	28.6	35.7	42.6	52.0	61.9	70.0	72.2	73.2	63.6	52.7	41.5	33.9
	DBStd	16.71	8.62	8.49	8.06	7.66	5.49	5.01	4.81	5.29	7.00	6.54	7.84	7.66
	HDD50	2203	664	403	246	68	2	0	0	0	1	48	272	499
	HDD65	5344	1129	821	693	392	123	13	5	6	109	384	705	964
	CDD50	3082	0	2	18	126	372	599	688	720	410	131	16	0
	CDD65	752	0	0	0	1	28	163	228	262	68	2	0	0
	CDH74	7928	0	0	1	64	339	1597	2329	2809	740	49	0	0
	CDH80	2983	0	0	0	4	49	548	887	1262	231	2	0	0
Wind	WSAvg	4.8	4.5	5.3	5.4	6.2	5.3	4.6	4.1	4.8	4.9	4.7	4.2	4.2
Precipitation	PrecAvg	24.4	1.5	1.1	1.5	2.2	2.8	3.1	3.0	1.8	2.2	2.0	1.5	1.7
	PrecMax	42.8	3.0	2.7	4.0	6.7	5.6	5.8	6.8	4.4	6.9	5.0	4.0	5.8
	PrecMin	11.8	0.2	0.1	0.1	0.7	0.5	0.9	0.7	0.3	0.1	0.0	0.0	0.0
	PrecStd	6.4	0.9	0.7	1.1	1.3	1.3	1.4	1.6	1.1	1.7	1.2	1.0	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	62.9	71.8	79.4	84.7	93.8	93.5	97.4	90.4	78.6	67.7	57.2
		MCWB	45.6	51.9	56.0	61.2	66.7	70.8	71.0	70.8	66.7	60.7	56.2	48.7
	2%	DB	46.6	56.5	66.2	75.3	80.7	89.1	90.8	92.6	86.0	74.3	61.3	51.8
		MCWB	42.4	48.9	52.9	60.0	64.6	69.7	70.2	68.3	64.7	59.5	53.9	46.5
	5%	DB	42.6	51.8	61.4	70.6	77.8	85.9	88.2	90.2	81.8	69.7	56.1	47.4
		MCWB	39.9	46.1	50.6	57.6	63.3	68.9	69.5	67.9	63.5	57.4	51.5	43.6
	10%	DB	39.5	47.6	56.7	66.6	74.6	82.9	85.3	87.2	77.5	65.4	52.3	43.8
		MCWB	37.5	43.4	47.9	55.3	61.8	67.8	68.8	67.5	62.5	56.1	49.3	41.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.7	53.8	57.2	62.8	68.4	74.2	74.5	74.3	69.3	62.3	57.6	49.7
		MCDB	52.0	61.4	70.2	76.2	81.7	88.1	87.3	89.0	83.8	72.9	65.1	53.5
	2%	WB	43.4	49.5	54.1	60.9	66.3	72.0	72.6	72.0	66.7	60.5	55.2	47.1
		MCDB	46.0	55.7	64.8	74.5	77.8	84.4	85.9	85.8	78.9	71.7	60.6	51.7
	5%	WB	40.6	46.7	51.2	58.9	64.5	70.5	71.3	70.5	65.2	58.8	52.5	44.4
		MCDB	42.9	51.2	60.3	69.8	75.2	82.5	83.8	83.9	77.2	68.1	55.8	47.8
	10%	WB	37.9	43.9	48.6	56.6	62.9	69.0	70.0	69.1	63.7	56.6	50.2	41.7
		MCDB	39.4	46.9	55.8	64.8	72.9	80.6	81.4	82.6	75.7	63.9	52.8	43.9
Mean Daily Temperature Range	5% DB	MDBR	13.4	14.8	17.6	20.0	20.4	21.7	22.7	24.1	21.5	19.3	13.2	12.7
		MCDBR	17.5	23.4	27.8	27.8	25.5	26.3	26.6	28.4	28.2	28.5	21.3	19.0
		MCWBR	12.6	16.2	16.0	14.5	11.3	9.8	8.9	8.2	10.1	14.7	13.2	14.0
	5% WB	MCDBR	16.1	22.7	25.2	25.0	23.9	24.0	23.3	24.2	23.6	25.3	18.5	19.6
		MCWBR	12.6	15.9	15.1	13.0	12.0	9.8	9.1	8.5	9.4	13.5	12.0	14.2
Clear-Sky Solar Irradiance	taub	0.332	0.363	0.399	0.457	0.434	0.439	0.450	0.469	0.427	0.389	0.364	0.336	
	taud	2.309	2.235	2.196	2.049	2.189	2.210	2.186	2.127	2.208	2.306	2.323	2.331	
	Ebn at Noon	243	255	262	255	266	265	260	249	250	244	229	229	
	Edh at Noon	26	34	40	50	45	44	45	46	39	31	26	23	
All-Sky Solar Radiation	RadAvg	449	717	1105	1498	1846	2027	2045	1803	1308	852	474	366	
	RadStd	66	112	123	147	141	144	134	119	106	109	61	54	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (71 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	-247	+108	N/A	

Nomenclature: See separate page